MIL-E-1/1713B
26 February 1979
SUPERSEDING
MIL-E-1/1713A
4 June 1976
(See note 13)

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, TRAVELING WAVE

1/ TYPE DOD-009

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Medium power amplifier, 4.4 to 5.0 GHz, periodic permanent magnet focused.

ABSOLUTE RATINGS:

Parameter:	Ef	If	Ec2	Ic2	Ew	Iw	Eb	Ib
Unit:	V	A	Vdc	mAdc	Vdc	mAdc	Vdc	mAdc
Maximum:	6.6	1.6	1,050	0.5	1,700	2.5	1,600	33
Minimum:	6.0	---	---	---	1,400	---	1,000	---

PHYSICAL CHARACTERISTICS:

Dimensions	See figure 1	RF connectors:	TNC, female (see figure 1)
Mounting position:	Any	Weight:	3 pounds (1.361 kg) maximum
Cooling:	Heat sink	Voltage reference:	Cathode
DC connectors:	See figure 1		

TEST CONDITIONS:

Parameter:	Ef	Ec2	Ew	Eb
Unit:	V	Vdc	Vdc	Vdc
	6.3	Set up from data on tube label and individual test data sheet (see notes 1 and 2)		

FREQUENCY	
F	GHz
1	4.4
2	4.5
3	4.6
4	4.7
5	4.8
6	4.9
7	5.0

GENERAL:

First article sample approval - Required (see note 3)

1/ Replaces tube types A1310V1 and VTC-6067B1

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS		UNIT
					MINIMUM	MAXIMUM	
	Quality conformance inspection, part 1	4					
4289	Heater current (nonoperating)	-	Ef = 6.3 V	If	0.2	1.6	A
1256	Electrode current (anode)	1,2		Ic2	---	0.5	mA
1261	Electrode voltage (anode)	1,2		Ec2	900	1,050	Vdc
1261	Electrode voltage (helix)	1,2		Ew	1,500	1,600	Vdc
1261	Electrode voltage (collector)	1,2		Eb	1,140	1,160	Vdc
1256	Electrode current (helix)	1,2	Pi = -5 to +2 dBm	Iw	---	2.5	mA
1256	Electrode current (collector)	1,2		Ib	---	33	mA
4251	Output power and gain variation	1,2,5	Pi(rf) = -5 dBm; F = F1 thru F7	ΔG	---	$1.25/_{100}$	dB/MHz
4250	Power output	1,2	Pi(rf) = -5 to +2 dBm; F = F1 thru F7	Po	1.8	4.5	W
4256	Input and output match (cold)	-	No voltages	VSWR	---	2:1	---
4261	Stability	1,6	F = F1 thru F7, no oscillation	---	---	---	---
	Quality conformance inspection, part 2						
4260	Noise figure	2,7	Pi(rf) = -15 dBm (max); F = F1 thru F7	NF	---	28	dB
4953	Small signal gain	---	Pi(rf) = -15 dBm	Gss	35	45	dB
1047	High- and low- temperature performance (operating)	8	T_A = -20° to +65°C	---	---	---	---
1105	Permanence of marking	-					
	Quality conformance inspection, part 3						
---	Life test	2	Group S; CW operations; Pi(rf) = +2 dBm; F1	t	2,500	---	hrs
---	Life-test end points:						
4250	Power output	2	Pi(rf) = -5 to +2 dBm; F = F1 thru F7	Po	1.0	4.5	W
1256	Electrode current (anode)	1,2		Ic2	---	0.5	mA
1256	Electrode current (helix)	1,2	Pi(rf) = -5 to +2 dBm; F = F1 thru F7	Iw	---	4.0	mA
1032	Vibration, mechanical	9,10,11	No voltages	---	---	---	---
1042	Shock, specified pulse	10,11		---	---	---	---
---	Heater voltage	11,12	Ef = 6.0 V	---	---	---	---
		11,12	Ef = 6.6 V	---	---	---	---

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NOTES:

1. A copy of the critical performance characteristics including operating voltages shall be shipped with each tube. The operating voltages (including Ew, Ib, Iw, Ec2, and Eb) shall be clearly indicated on the tube label and on a small decal at the collector end of each tube. Each tube delivered on the contract shall operate optimally under a single combination of discrete voltages which fall within the ranges shown under absolute ratings. All operating requirements of this specification shall be met under these specific voltages.

2. With no signal applied, apply Ef = 6.3 V for 180 seconds (min). Set collector, helix, and anode voltages to the optimum values shown on the TUT decal.

3. First article sample inspection shall conform to the requirements of appendix F of MIL-E-1 and shall consist of performing all tests specified on this tube specification sheet. Invitation for bids should provide that the preparing activity reserves the right to waive the requirements for first article samples as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

4. Unless otherwise specified, the AQL for all tests listed under quality conformance inspection, part 1, shall be 1.0, inspection level II.

5. This gain variation shall be determined by means of a swept signal technique, or equivalent method. If a swept signal technique is used, the sweep rate shall be such as to assure that the gain at each point is identical to that measured on a single frequency basis. If a single frequency method is used, gain measurements shall be made at frequency increments of no greater than 100 MHz.

6. The TUT input coupler shall be terminated with a matched load. The output shall be terminated into a 4:1 mismatch varied over 180 degrees.

7. An alternate noise figure test may be used provided all requirements specified herein are satisfied in accordance with the requirements of The Alternate test methods paragraph of MIL-STD-1311.

8. The tube shall be capable of meeting all quality conformance inspection, part 1, tests when operated at the following stabilized temperatures.

Temperature	Method	Test
-20° to +65°C	4250 1256 4261	Power output Electrode current (helix) Stability
+25°C	4250	Power output

9. The tube shall be mounted in a resonance free jig and vibrated with sinusoidal excitation in each of three mutually perpendicular planes through the following amplitudes:

10 to 100 Hz	10 G (or 0.1 inch D.A. max)
100 to 500 Hz	3G

Cycling test: The frequency shall vary from 10 to 500 to 10 Hz with approximately logarithmic progression, and shall require approximately 30 minutes to traverse the range. This constitutes one cycle. The TUT shall be vibrated for one such cycle in each of the three planes..

Resonance test: Mechanical resonant frequencies of the tube shall be determined during the cycling test.

Vibration test schedule (times shown refer to one axis)

Type	Room temperature
Resonance	30 minutes
Cycling	30 minutes

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NOTES: -Continued

10. After performance of the specified test, the tube shall be capable of satisfying the requirements of the following quality conformance inspection, part 1 tests; power output, electrode current (helix), and stability.

11. This test shall be performed during the initial production and once each succeeding 12-calendar months in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of one. In the event of failure, the test will be made as a part of quality conformance inspection, part 2, code level D, with an AQL of 6.5. The regular "12-calendar month" double sampling plan shall be reinstated after three consecutive samples have been accepted.

12. In addition, the TUT shall satisfy all tests specified in QCI, part 1 at Ef = 6.0 V at Ef = 6.6 V.

13. Revision letters are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Air Force - 99
DLA - ES

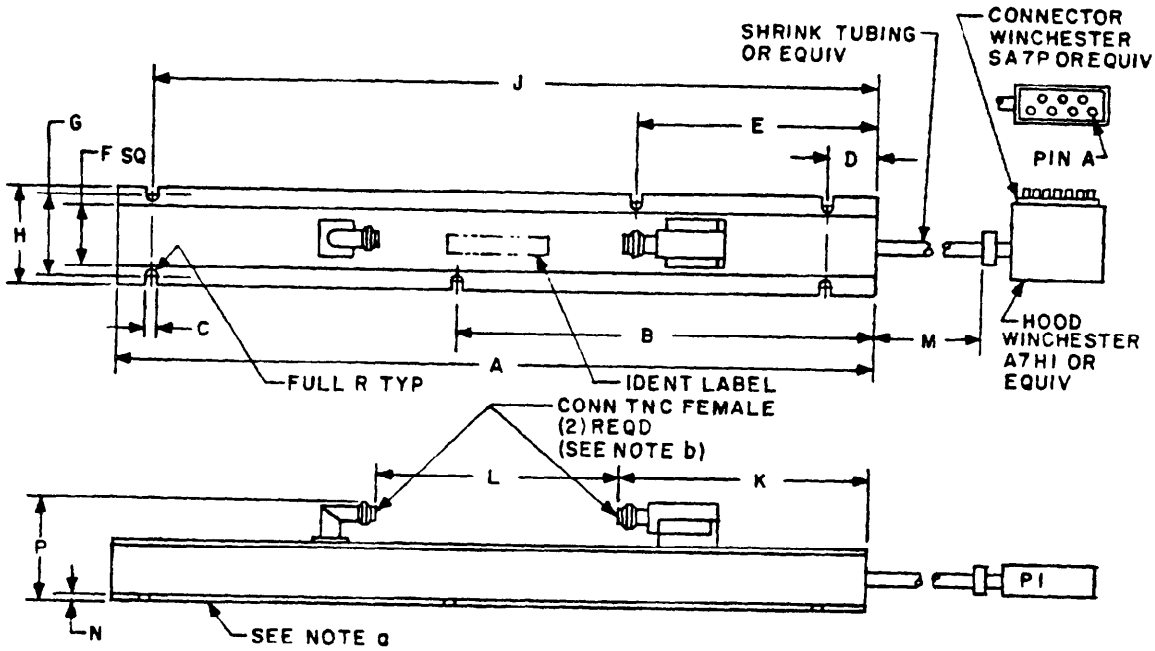
User activities:
Army - AR, WC
Navy - AS, OS, MC, CG
Air Force - 11

Preparing activity:
Navy - EC

Agent:
DLA - ES

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Color	Element	Pin
	Spare	A
Brown	Heater	B
Yellow	Heater Cathode	C
	Spare	D
Blue	Anode	E
Orange	Helix	F
Black	Collector/Capsule	H

Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 1 (see note 4)		
A		15.38 (390.65)
B	8.241(209.32)	8.259(209.78)
C	.172 (4.37)	.202 (5.13)
D	.985 (25.02)	1.015 (25.78)
E	4.735(120.27)	4.765(121.03)
F		1.250 (31.75)
G		1.500 (38.10)
H		1.94 (49.3)
J	14.297(363.14)	14.327(363.91)
M	10.75 (273.0)	11.25 (285.8)
N	.125 (3.18)	.155 (3.94)
Quality conformance inspection, part 2		
K	5.25 (133.4)	5.75 (146.0)
L	5.12 (130.0)	5.62 (142.7)
P		2.00 (50.8)

NOTES:
a. This surface is to be finished per MIL-E-5400 and MIL-C-5541 as applicable to provide a conductive protective finish (Chromate conversion).
b. Clearance shall be maintained between the connector and the tube body to allow for .640 (16.26 mm) DIA mating connector.

FIGURE 1. Outline drawing of electronic tube type DOD-009.

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DOCUMENT IDENTIFIER *(Number)* AND TITLE MIL-E-1/1713B, Electron Tube, Traveling Wave 1/ Type DoD-009

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